

2.10 Applying Predicate Logic to Real-World Problems

Predicate logic can be used to solve real-world problems by representing knowledge clearly and systematically. It helps model situations involving objects, their properties, and relationships.

DO YOU KNOW? Predicate logic can model family trees, social networks, and even computer programs!

Representing Relationships Using Predicates

Examples:

- $\text{Borrows}(x, y) \rightarrow \text{Student } x \text{ borrows book } y$
- $\text{Teaches}(t, s) \rightarrow \text{Teacher } t \text{ teaches student } s$

These predicates clearly describe how objects are connected.

Writing Logical Statements With Quantifiers

Quantifiers allow us to describe both general situations and specific cases in the library system. Examples from a library system:

- $\forall x, \text{Student}(x) \rightarrow \text{Borrows}(x, \text{Book}) \rightarrow \text{"Every student borrows a book"}$
- $\exists y, \text{Book}(y) \wedge \text{Available}(y) \rightarrow \text{"There exists at least one book that is available"}$

Reasoning and Drawing Conclusions

Using predicate logic, we can draw conclusions from given facts.

Example: If $\forall x, \text{Student}(x) \rightarrow \text{HasLibraryCard}(x)$ and "Ali is a student" then we conclude "Ali has a library card."

This helps in decision making and ensures correct and consistent conclusions.

Short Questions

1. How can predicate logic be applied to real-world problems?
 2. Give two examples of predicates used to represent relationships.
 3. How do quantifiers help in writing rules for real-world systems?
 4. Give an example of a logical statement using quantifiers in a library system.
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